

MA341

APPLIED ODE'S

Review 2

1. Find $|A|$, A^{-1} , and eigenvalues and eigenvectors of the following matrices

a. $\begin{pmatrix} 5 & 1 \\ 2 & 4 \end{pmatrix}$

Solution: $\det A = 18$

$$A^{-1} = \frac{1}{18} \begin{pmatrix} 4 & -1 \\ -2 & 5 \end{pmatrix}$$

Eigenvalues: $\det(A - \lambda I) = \lambda^2 - 9\lambda + 18 = (\lambda - 3)(\lambda - 6)$. Hence $\lambda_1 = 3, \lambda_2 = 6$.

Eigenvectors: $v_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$.

b. $\begin{pmatrix} 2 & 2 & -1 \\ 1 & 3 & -1 \\ -1 & -2 & 2 \end{pmatrix}$

Solution:

$$\det A = 5$$

$$A^{-1} = \begin{pmatrix} \frac{4}{5} & -\frac{2}{5} & \frac{1}{5} \\ -\frac{1}{5} & \frac{3}{5} & \frac{1}{5} \\ \frac{1}{5} & \frac{2}{5} & \frac{4}{5} \end{pmatrix}$$

Eigenvalues:

$$\det(A - \lambda I) = -\lambda^3 + 7\lambda^2 - 11\lambda + 5 = -(\lambda - 5)(\lambda - 1)^2$$

Hence $\lambda_{1,2} = 1, \lambda_3 = 5$.

Eigenvectors $v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, v_2 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, v_3 = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$

c. $\begin{pmatrix} -1 & 0 & 1 \\ -3 & 4 & 1 \\ 0 & 0 & 2 \end{pmatrix}$

Solution:

$$\det A = -8$$

$$A^{-1} = \begin{pmatrix} -1 & 0 & \frac{1}{2} \\ -\frac{3}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 0 & \frac{1}{2} \end{pmatrix}$$

Eigenvalues:

$$\det(A - \lambda I) = -\lambda^3 + 5\lambda^2 - 2\lambda - 8 = -(\lambda + 1)(\lambda - 2)(\lambda - 4)$$

Hence $\lambda_1 = -1$, $\lambda_2 = 2$, $\lambda_3 = 4$.

$$\text{Eigenvectors } v_1 = \begin{pmatrix} -5 \\ 3 \\ 0 \end{pmatrix}, v_2 = \begin{pmatrix} 2 \\ 0 \\ 6 \end{pmatrix}, v_3 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

2. Solve the following homogeneous problem $x'(t) = \begin{pmatrix} 3 & -2 \\ 4 & -1 \end{pmatrix} x(t)$

Solution:

$$x(t) = c_1 e^t \begin{pmatrix} \cos 2t \\ \cos 2t + \sin 2t \end{pmatrix} + c_2 e^t \begin{pmatrix} \sin 2t \\ \sin 2t - \cos 2t \end{pmatrix}$$

3. Solve the following systems of ODEs subject to given initial conditions.

a. $x'(t) = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} x(t) + \begin{pmatrix} e^t \\ t \end{pmatrix}, x(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

Solution: General solution

$$x(t) = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t + \begin{pmatrix} t + \frac{1}{4}e^t(6t - 1) \\ \frac{1}{4}(3e^t + 4)(2t - 1) \end{pmatrix}$$

Use IC

$$x(0) = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} \frac{1}{4}(-1) \\ \frac{1}{4}(3 + 4)(-1) \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} - \frac{1}{4} \begin{pmatrix} 1 \\ 7 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

to find $c_1 = \frac{5}{4}$, $c_2 = 0$. Thus,

$$x(t) = \frac{5}{4} \begin{pmatrix} 1 \\ 3 \end{pmatrix} e^{-t} + \begin{pmatrix} t + \frac{1}{4}e^t(6t - 1) \\ \frac{1}{4}(3e^t + 4)(2t - 1) \end{pmatrix}$$

b. $x'(t) = \begin{pmatrix} 5 & 3 \\ -6 & -4 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ e^t \end{pmatrix}, x(0) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

Solution: General solution

$$x(t) = c_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{2t} + c_2 \begin{pmatrix} -1 \\ 2 \end{pmatrix} e^{-t} + \begin{pmatrix} -\frac{3e^t}{2} - 2 \\ 2e^t + 3 \end{pmatrix}$$

Use IC

$$x(0) = c_1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} -1 \\ 2 \end{pmatrix} + \begin{pmatrix} -\frac{3}{2} - 2 \\ 2 + 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

to find $c_1 = -4, c_2 = -\frac{1}{2}$. Thus,

$$x(t) = -4 \begin{pmatrix} -1 \\ 1 \end{pmatrix} e^{2t} - \frac{1}{2} \begin{pmatrix} -1 \\ 2 \end{pmatrix} e^{-t} + \begin{pmatrix} -\frac{3e^t}{2} - 2 \\ 2e^t + 3 \end{pmatrix}$$

c. $x'(t) = \begin{pmatrix} 2 & -1 \\ 5 & -2 \end{pmatrix} x(t) + \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}, x(0) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

Solution: General solution

$$x(t) = c_1 \begin{pmatrix} 2 \cos t - \sin t \\ 5 \cos t \end{pmatrix} + c_2 \begin{pmatrix} 2 \sin t + \cos t \\ 5 \sin t \end{pmatrix} + \begin{pmatrix} (t - \frac{1}{5}) \sin t + (t + \frac{9}{10}) \cos t \\ \frac{1}{2}(6t - 1) \sin t + (t + 2) \cos t \end{pmatrix}$$

Use IC

$$x(0) = c_1 \begin{pmatrix} 2 \\ 5 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} \frac{9}{10} \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

to solve for $c_1 = -\frac{1}{5}, c_2 = -\frac{1}{2}$. Thus,

$$x(t) = -\frac{1}{5} \begin{pmatrix} 2 \cos t - \sin t \\ 5 \cos t \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 2 \sin t + \cos t \\ 5 \sin t \end{pmatrix} + \begin{pmatrix} (t - \frac{1}{5}) \sin t + (t + \frac{9}{10}) \cos t \\ \frac{1}{2}(6t - 1) \sin t + (t + 2) \cos t \end{pmatrix}$$